

RESEARCH SUMMARY

Highlights	ICML 2026 paper on masked diffusion LMs: G-Star (poster) Co-supervising 3 MSc theses as a 1st year PhD student IOI national team candidate; 8th in the national Mathematics olympiad out of 1M participants	
Topics	Efficient LLM inference	Discrete Diffusion, Flow Maps, Consistency, Inference-Time Scaling, Speculative Decoding
Skills	Libraries	PyTorch, Hugging Face Transformers (Qwen, LLaMA)
	Training / inference	Multi-node SLURM (DDP/FSDP), DeepSpeed, FlashAttention, bf16, vLLM/SGLang
	On-device inference	Pruning, quantization, ONNX
Background	5+ years in ML (engineer → researcher), PhD student; available from Jun 2027	

PUBLICATIONS

Guided Star-Shaped Masked Diffusion — V. Meshchaninov, E. Shibaev, A. Makoian, et al.	ICML 2026, poster
• Fine-tuning pretrained masked diffusion LMs for guided correction of already generated tokens	
Motion ID: Human Authentication Approach — A. Gavron, et al., A. Makoian	arXiv 2023
• User authentication by motion patterns: 82.2% ± 9% accuracy (baseline 70%)	

RESEARCH EXPERIENCE

JetBrains	Feb 2025 — Present
ML Researcher / JetBrains Research <i>with D. Vetrov</i>	
Python 3, PyTorch, JAX, transformers, Hydra, Lightning, SLURM	
→Masked Diffusion LMs	
• Inference-time scaling, diffusion drafters for speculative decoding, curriculum learning (Dream-7B, Qwen3-0.6B MDLM/BD3LM)	
→Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference	
• Distillation (flow maps, consistency) and latent space compression for real-time acceleration	
AIRI (Artificial Intelligence Research Institute)	Jul 2022 — Oct 2022
ML Research Intern / GAN adaptation	
Python 3, PyTorch, Hydra, WandB	
• Few-Shot Image Generation implementation for few-shot GAN adaptation to a new domain from ~10 images	
Yandex	Jul 2021 — Oct 2021
ML Researcher / VR Team <i>with V. Lempitsky</i>	
Python 3, PyTorch	
• Trimap-free human alpha matting for VR scenes: improved FBA Matting , built alpha-mask datasets	
Earlier: Samsung R&D Department , ML Research Intern, published Motion ID	Nov 2020 — Jun 2021

EDUCATION

PhD in Computer Science (1st year) , Constructor University , Bremen, Germany	Sep 2026 — Present
• Teaching Assistant (seminars): Deep Bayesian Models (Spring 2026), Bayesian Methods (Fall 2026)	
• Co-supervising 3 MSc theses: Loop Transformers, Gaussian Diffusion LMs, Jacobi Decoding	
Master in Computer Science , Constructor University , Bremen, Germany	Sep 2024 — May 2026
Bachelor in Computer Science , Neapolis University of Paphos , Cyprus, GPA 86/100	Jan 2023 — Jun 2024
Bachelor in Computer Science , HSE University , Moscow, Russia, GPA 8.7/10	Sep 2019 — Jun 2024
Master's Equivalent in Data Science , Yandex School of Data Analysis , Russia	Sep 2020 — May 2022

INDUSTRY EXPERIENCE

Yandex	Jun 2023 — Sep 2024
Middle ML Engineer / Geo services	
Python 3, PyTorch, Hydra, OpenCV, LangChain	
• Road segmentation from aerial imagery with IoU 0.93; post-processing reduced artifacts from 10% → 0.02%	
• News classification pipeline with LangChain agents for map update hypotheses: precision 0.9, recall 0.85	
Sber Devices	Jul 2022 — Jun 2023
Middle ML Engineer / Computer Vision Platform	
Python 3, C++, PyTorch, ONNX, torch_pruning	
• Real-time background removal for video conferencing (0.98 Dice); pruning gave 1.5x speed-up	
• On-device 700 KB payment card segmentation model with C++ inference in a multiplatform mobile SDK	

COMPETITIONS

Candidate for the national team at the IOI (International Olympiad in Informatics)	May, Nov 2018
Winner of the national olympiad in Mathematics (8 th) and Informatics (33 rd) out of 1M participants	Apr 2018, 2019
Top 8% in Google Hash Code. Top 50 in Google Kick Start. Top 350 (top 0.8%) in Google Code Jam	Feb, May, June 2020